

Cross-Domain Large Margin Distribution Machine

Junlin Chen^a, Jinrui Guo^a, Wenting Hu^b, Wenzhe Pan^a and Guojun Huang^{a,*}

^aSchool of Artificial Intelligence, Southwest University, Chongqing 400715, China

^bBusiness school, Jiangsu open University, Jiangsu 210036, China

ARTICLE INFO

Keywords:

Electronic noses

Large margin distribution machine

Cross domain

Sensor drift

ABSTRACT

Due to sensor drift issues, electronic noses (E-noses) are severely limited in their applications. Support vector machine (SVM) aims to reduce classification errors by maximizing the minimum margin, which is widely used to classify drift-compensated datasets in E-noses. However, the cross-domain learning ability of SVM remains constrained, leading to poor performance in addressing the sensor drift in E-noses. Moreover, SVM exhibits limitations in terms of generalization, and recent theoretical research has revealed that optimizing the margin distribution result in more superior generalization capabilities than maximizing the minimum margin. To address the aforementioned shortcomings, this paper proposes a novel cross-domain large margin distribution machine (CDLDM), which achieves excellent cross-domain learning capabilities and high generalization performance. Specifically, CDLDM aims to minimize the distribution discrepancy between source and target domains by simultaneously optimizing the projected maximum marginal distribution discrepancy, projected maximum distribution scattering discrepancy, and the manifold consistency under marginal distribution, thus achieving effective domain adaptation. In addition, in order to utilize the distribution characteristic information of the samples, the mean and variance optimization terms of the margin distribution are added to the objective function, which further improves the generalization ability of CDLDM. Finally, experiments conducted on the E-noses datasets have demonstrated the significant advantages of CDLDM in the field of E-noses.

1. Introduction

Electronic noses (E-noses) consists of a cross sensitive chemical sensor array and a pattern recognition algorithm. Currently, E-noses are widely employed in diverse domains such as medical diagnostics (Dragonieri et al., 2009, 2007), food analysis (Di Natale et al., 1997; Pardo et al., 2000), and air quality monitoring (Dutta et al., 2003; Jiang et al., 2019). In practical applications, the discriminative capability of E-noses can be compromised by factors such as sensor aging, chemical contamination, and environmental fluctuations, which can induce sensor drift (Hossein-Babaei & Ghafarinia, 2010; Zhang et al., 2013; Güney & Atasoy, 2015). Sensor drift seriously hinders the recognition accuracy of E-noses in practical scenarios Marco & Gutierrez-Galvez (2012). As is well known, the drift of E-noses caused by gas sensors can lead to inconsistent distributions of training data and test data, thus causing cross-domain problems Zhang & Zhang (2015). In order to facilitate comprehension, we offer a succinct explanation of cross-domain issues. Let $X_1, \dots, X_n$ represent the source domain, and the distribution of the source domain is represented by p_1 . $Y_1, \dots, Y_n$ represent the target domain and p_2 represents its distribution. The drift issue refers to the fact that the distributions of two domains are inconsistent, i.e., $p_1 \neq p_2$. In this scenario, an E-nose classifier trained on data from the source domain often exhibits significantly reduced performance when tested on data from the target domain Di Carlo et al. (2012). As the aging of sensor increases, the discrepancy between the distributions of $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ grows more pronounced, leading to a degradation in the classifier's generalization capabilities.

To address the issue of sensor drift in E-noses, a prevalent method is compensating for the drift data to ensure the alignment of the source and target domain distributions. Subsequently, a support vector machine (SVM) classifier is employed to classify (Yan et al., 2022; Zhang et al., 2017; Guo et al., 2022). While this strategy often yields favorable outcomes, it is also with potential drawbacks. SVM operates under the assumption of identical data distribution between the training and testing datasets, i.e., $p_1 = p_2$. Consequently, it faces limitations in solving cross-domain problems.

*Corresponding author: Guojun Huang

E-mail address: jlchen08@163.com (J. Chen),

jrguo13@163.com (J. Guo), xiaohu_nanjing@163.com (W. Hu),

panwz321@outlook.com (W. Pan), aikyoko@163.com (G. Huang)

ORCID(s):

Despite compensating for data drift, SVM neglects distribution discrepancy between various domains during training, potentially hindering optimal performance in certain scenarios.

At present, some domain adaptation methods based on SVM have been proposed to solve cross-domain problems. Long et al. (Long et al., 2014a) proposed a new transfer learning framework called adaptive regularization based transfer learning (ARTL). ARTL learns an adaptive classifier by simultaneously optimizing the structural risk function, joint distribution matching between domains, and manifold consistency under marginal distribution. Under the ARTL framework, Long et al. proposed ARSVM (Long et al., 2014b) and derived corresponding solutions using the representer theorem in reproducing kernel Hilbert spaces. Tao et al. (Tao et al., 2012) have developed the generalized projection maximum distribution difference (GPMDD) measure, which is grounded in the reproducing kernel Hilbert space (RKHS) domain distribution. This measure takes into account both the mean and scattering differences of the maximum projection distributions, thereby providing a nuanced assessment of the distributional variations. Based on the principles of structural risk and GPMDD minimization, Tao et al. (Tao et al., 2012) proposed a new domain adaptive kernel support vector machine (DAKSVM). The above models have achieved good cross-domain processing performance to a certain extent, but they only adopt some cross-domain processing methods, ignoring some important methods. In addition, these models are all improved under the SVM model and still lack consideration for marginal distribution.

SVM aims to maximize the minimum margin between the support vector and the classification hyperplane (Reyzin & Schapire, 2006; Zhang & Zhou, 2018; Liu et al., 2021). However, according to the marginal theory, the marginal distribution plays a more crucial role than the minimum margin in enhancing the model's generalization capability (Gao & Zhou, 2013; Wang et al., 2011). Inspired by these results, Zhang et al. Zhang & Zhou (2014a) first focused on the influence of marginal distribution on SVM and proposed the large margin distribution machine (LDM). By incorporating two second-order statistical measures, the construction of the classification hyperplane is affected by all sample points rather than being confined to the support vectors alone (Sain, 1996). Hence, this comprehensive consideration of data points is posited to yield superior generalization performance in comparison to the traditional SVM methodology. Despite its widespread application, LDM has predominantly been employed in addressing single-domain problems and exhibits limited efficacy in resolving cross-domain challenges.

Inspired by previous researches, we extend LDM to the cross-domain learning domain by considering the global distribution differences of data and propose a novel cross-domain large margin distribution machine (CDLDM) with excellent cross-domain learning capabilities and high generalization performance. Specifically, CDLDM aims to minimize the distribution difference between the source domain and the target domain by simultaneously optimizing the projected maximum marginal distribution difference, the projected maximum distribution dispersion difference, and the manifold consistency under the marginal distribution, thus achieving cross-domain learning. In addition, CDLDM also optimizes marginal distribution and demonstrates superior generalization performance than SVM. Compared with the previous method of compensating for drift data before applying SVM classification, CDLDM extends SVM to cross-domain learning domains, enabling it to directly classify drift data and possessing excellent generalization ability. We conduct a sequence of comprehensive experiments on different sensor drift datasets to empirically verify the cross-domain learning capability of CDLDM. The principal contributions of this paper are synthesized in the following key points:

- To the best of our knowledge, this paper extends LDM to the field of cross-domain learning for the first time, greatly enhancing the cross-domain learning capabilities of LDM.
- From the perspective of machine learning, we provide a new and efficient solution to the sensor drift problem. The proposed CDLDM is a domain adaptive classifier that has cross-domain capabilities and effectively solves the problem of distribution inconsistency caused by drift.
- Based on a series of comprehensive experiments on different E-noses datasets, consistent findings highlight the excellent cross-domain learning capabilities exhibited by CDLDM. Notably, our proposed model outperforms some classical models.

The structure of the remaining sections of this manuscript is outlined below. Section 2 delineates a succinct overview of the literature on drift compensation techniques for gas sensor, while concurrently introducing the LDM. Section 3 proposes the CDLDM, along with its algorithmic solution. In Section 4, an extensive experimental protocol is elaborated, wherein a diverse array of sensor drift datasets is utilized to demonstrate the enhanced robustness against drift of the proposed model. Conclusively, Section 5 synthesizes the findings of this study.

2. Related Work

In this section, we briefly introduce some background knowledge of sensor drift compensation and LDM.

2.1. Drift Compensation for Gas Sensors

In contemporary research, the resolution of the drift issues is principally stratified into three approaches: 1. drift component correction methods; 2. adaptive methods; 3. machine learning techniques. Drift component correction strategies aim to identify and eliminate inherent drift components in original signals before classifier training. This methodology relies on reference samples or component removal techniques. Haugen et al. (Haugen et al., 2000) introduced an algorithmic framework for mathematical drift compensation, which is predicated on the utilization of correction samples to achieve enhanced calibration accuracy, yet it imposes stringent demands on calibration samples, requiring a high correlation with the target sample and comparable sensor responses and drift behaviors. Another approach, the orthogonal signal correction (OSC) model proposed by Padilla et al. (Padilla et al., 2010) identifies and removes drift by targeting components orthogonal to the signal.

Adaptive compensation is a method to address drift by dynamically adjusting coefficients in pattern recognition algorithms to align with current sensor outputs. This proves effective for tackling time-varying drift challenges. Proposed strategies, such as Adaptive Resonance Theory (Distante et al., 2000) and Self-Organizing Maps (SOM) (Zuppa et al., 2004), exhibit notable effectiveness in drift suppression. However, their success relies heavily on accurate model classification, with misclassifications potentially compromising the model's ability to track patterns accurately. Inspired by Martinelli et al.'s (Martinelli et al., 2013) artificial immune system, an adaptive classifier was integrated to mitigate drift impacts. Despite these advancements, their reliance on precise model classification and the need for frequent recalibration in practical scenarios pose inherent limitations, resulting in increased implementation costs and system complexity.

Machine learning methodologies are engineered for mitigating the impact of data drift or shift, with an emphasis on minimizing these effects without explicitly characterizing the nature of the drift or shift. Machine learning methodologies are classified into two main classes: decision-based and feature-based. Decision-based strategies aim to directly enhance the classifier's adaptability to drift (Liu et al., 2019; Vergara et al., 2012a). In contrast, feature-based strategies first compensate for drift data in the feature subspace to ensure the alignment of the source and target domain distributions. Subsequently, traditional machine learning models are applied in this subspace for classification (Yan et al., 2022; Zhang et al., 2017; Guo et al., 2022).

LDM promotes the generalization ability of traditional SVM by optimizing the margin distribution, diverging from the exclusive focus on minimizing the maximum margin. Despite its widespread application, LDM has predominantly been employed in addressing single-domain problems and exhibits limited efficacy in resolving cross-domain challenges. To address drift issues in E-noses data more effectively, we extend LDM to the field of cross-domain learning and proposed a novel CDLDM. In contrast to previous methodologies, which initially seek to align the distributions of original and drifted data in a subspace before applying machine learning techniques for classification, our approach simultaneously minimizes distribution differences between domains and classifies. This novel approach aims to offer a more efficient solution for mitigating drift-related challenges in E-noses systems.

2.2. Large Margin Distribution Machine

Let $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$ denote the training sample set, where $\mathbf{x}_i \in \mathbb{R}^n$, $y_i \in \{-1, +1\}$. The input data matrix denoted as $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_m]$ and the vector of sample labels is represented by $\mathbf{Y} = [y_1, \dots, y_m]^T$. The mapping of the feature vector $\mathbf{x}_i$ is represented by the function $\phi(\mathbf{x}_i)$. The matrix $\phi(\mathbf{X}) = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_m)]$ represents a high-dimensional set of samples. LDM aims to learn a linear function $f(\mathbf{x}) = \boldsymbol{\omega}^T \phi(\mathbf{x})$ with strong generalization performance to better predict target examples, where $\boldsymbol{\omega}$ is a weight vector.

LDM adopts a better marginal distribution than the minimum margin as the optimization objective Gao & Zhou (2013). The marginal distribution has first-order and second-order statistics. The first-order margin mean is

$$\bar{\gamma} = \frac{1}{m} \sum_{i=1}^m y_i \boldsymbol{\omega}^T \phi(\mathbf{x}_i) = \frac{1}{m} (\phi(\mathbf{X}) \mathbf{Y})^T \boldsymbol{\omega}, \quad (1)$$

and the second-order variance of margin is

$$\begin{aligned}\hat{\gamma} &= \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m (y_i \omega^T \phi(\mathbf{x}_i) - y_j \omega^T \phi(\mathbf{x}_j))^2 \\ &= \frac{2}{m^2} (m \omega^T \phi(\mathbf{X}) \phi(\mathbf{X})^T \omega - \omega^T \phi(\mathbf{X}) \mathbf{Y} \mathbf{Y}^T \phi(\mathbf{X})^T \omega).\end{aligned}\quad (2)$$

LDM optimizes the margin distribution by simultaneously maximizing margin mean and minimizing margin variance, which are represented as

$$\begin{aligned}\min_{\omega, \xi} \quad & \frac{1}{2} \omega^T \omega + \lambda_1 \hat{\gamma} - \lambda_2 \bar{\gamma} + C \sum_{i=1}^m \xi_i, \\ \text{s.t.} \quad & y_i \omega^T \phi(\mathbf{x}_i) \geq 1 - \xi_i, \\ & \xi_i \geq 0, \quad i = 1, 2, \dots, m,\end{aligned}\quad (3)$$

where $C, \lambda_1, \lambda_2 \geq 0$ are the parameters for trading-off the training error, the margin variance, the margin mean and the model complexity.

3. Proposed Methodology

In this section, we first provide a notation explanation in Section 3.1. Then, a novel efficient Cross-Domain Large Margin Distribution Machine is presented in Section 3.2. As shown in Fig. 1, the proposed CDLDM is intended minimize the global distribution discrepancy of data. Finally, the pseudocode for the CDLDM algorithm is outlined in Section 3.3.

3.1. Notations

In the field of machine learning, a domain $\mathcal{D}$ consists of a feature space $\mathbf{X}$ and a marginal probabilistic distribution $P(\mathbf{X})$ on the feature space, i.e., $\mathcal{D} = \{\mathbf{X}, P(\mathbf{X})\}$, where $\mathbf{X} = \mathbf{x}_{i=1}^m \in \mathbf{X}$. After giving a domain $\mathcal{D} = \{\mathbf{X}, P(\mathbf{X})\}$, a task $\mathcal{T}$ consists of a label $\mathbf{Y}$ and an objective prediction function $f(\cdot)$, represented as $\mathcal{T} = \{\mathbf{Y}, f(\cdot)\}$. The objective prediction function $f(\cdot)$ cannot be directly observed, but can be learned from training data composed of feature label pairs $(\mathbf{x}_i \in \mathbf{X}, y_i \in Y)$. From the perspective of probability theory, the objective prediction function $f(\cdot)$ can be expressed as $P(Y|\mathbf{X})$. The task is represented as $\mathcal{T} = \{Y, P(Y|\mathbf{X})\}$. In the problem of E-nose drift, there are two crucial concepts: The source domain represents as $\mathcal{D}_S = \{(\mathbf{X}_S^1, y_S^1), \dots, (\mathbf{X}_S^m, y_S^m)\}$ and the target domain represents $\mathcal{D}_T = \{(\mathbf{x}_T^1, y_T^1), \dots, (\mathbf{x}_T^n, y_T^n)\}$. The data of the source and target domains are represented as $\mathbf{X}_S = [\mathbf{X}_S^1, \dots, \mathbf{X}_S^m] \in \mathcal{R}^{D \times N_S}$ with corresponding labels $\mathbf{Y}_S = [y_S^1, \dots, y_S^n] \in \mathcal{R}^{D \times C}$ and $\mathbf{X}_T = [\mathbf{x}_T^1, \dots, \mathbf{x}_T^n] \in \mathcal{R}^{D \times N_T}$ with corresponding labels $\mathbf{Y}_T = [y_T^1, \dots, y_T^n] \in \mathcal{R}^{D \times C}$, where D represents the feature dimensionality, and C indicates the overall count of categories. $N_S = m$ denotes the number of samples in the source domain, while in the target domain, it is represented as $N_T = n$. Although the source and target domains typically pertain to the same type of task, their data distributions exhibit distinct characteristics, i.e., $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$.

3.2. Cross-Domain LDM

In this section, we elaborate on the methodology for constructing the model, as detailed in Section 3.2.1. The measure of the projection distribution discrepancy is shown in detail in Section 3.2.2. Finally, Section 3.2.3 presents the optimization procedure.

3.2.1. Model construction of CDLDM

LDM is a widely used algorithm. Its core idea is to obtain better generalization performance by optimizing the margin distribution. For a hyperplane classifier like LDM, the decision function is expressed as $f(\mathbf{x}) = \omega^T \mathbf{x} + b$, where ω is a normal projection vector and b is the offset. Conventional machine learning paradigms typically presuppose that the training and test datasets adhere to an identical distribution, i.e., $P_S(\mathbf{X}_S, \mathbf{Y}_S) = P_T(\mathbf{X}_T, \mathbf{Y}_T)$. In this case, traditional LDM performs well. However, in many cases, this same distribution assumption is not satisfied, i.e., $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$. The drift caused by gas sensors in E-nose systems is precisely such a situation. Hence, in addressing the challenge of E-noses drift, our objective is to identify a feature transformation, $f(\mathbf{x}) = \omega^T \phi(\mathbf{x})$,

within a universal RKHS that effectively reduce the discrepancy between the data distributions of the two domains, thereby achieving cross-domain learning. Drawing inspiration from the concept of Manifold Regularization (Belkin et al., 2006), we can formulate the optimization problem of CDLDM as follows

$$f = \min_{\omega \in H_K} C \sum_{i=1}^m V(\mathbf{x}_i, y_i, f) + \frac{1}{2} \|f\|_K^2 + \lambda_1 \hat{\gamma} - \lambda_2 \bar{\gamma} + \lambda D_g(\mathbf{X}_S, \mathbf{X}_T), \quad (4)$$

where V represents the risk function. K refers to a kernel with its associated mapping function. $\|f\|_K^2$ is the L_2 norm of the function f . The loss function adopts the hinge loss function used by traditional LDM, which is represented as $V = (1 - y_i f(\mathbf{x}_i))_+$, where $(x)_+ = x$ if $x \geq 0$ and 0 otherwise. Hence, Eq. (4) can be further represented by the given function $f(\mathbf{x}) = \langle \omega, \phi(\mathbf{x}) \rangle_H$ as

$$f = \min_{\omega \in H_K} \frac{1}{2} \omega^T \omega + \lambda_1 \hat{\gamma} - \lambda_2 \bar{\gamma} + C \sum_{i=1}^m \xi_i + \lambda D_g(\mathbf{X}_S, \mathbf{X}_T), \quad (5)$$

$$s.t. \ y_i \omega^T \phi(\mathbf{x}_i) \geq 1 - \xi_i, \\ \xi_i \geq 0, \quad i = 1, 2, \dots, m,$$

where $\mathbf{x}_i \in \mathbf{X}_S$ and H represents the complete inner product space for f . And $D_g(\mathbf{X}_S, \mathbf{X}_T)$ is a measure of the projection distribution discrepancy, specifically represented as follows

$$D_g(\mathbf{X}_S, \mathbf{X}_T) = \lambda_3 D_f + \lambda_4 S_f + \lambda_5 P_f, \quad (6)$$

where, D_f denotes the discrepancy in the projection marginal distribution of RKHS embedding domains, S_f represents the discrepancy in the projection scattering distribution on these domains, and P_f refers to the graph-based Laplacian regularization.

Remark 1. The CDLDM is a universal framework that can degenerate into conventional models as needed.

- When $\lambda = 0$, the CDLDM model reverts to LDM. This study marks the first instance of extending the LDM model to the domain of cross-domain learning, addressing the real-world challenge of knowledge transfer across multiple domains. While traditional LDM primarily focuses on learning within a single domain, practical applications frequently involve the problem of knowledge transfer between multiple domains, i.e., how to effectively apply knowledge and experience acquired in one domain to others. The introduction of the CDLDM model aims to tackle this complex challenge.
- When $\lambda, \lambda_1, \lambda_2 = 0$, the CDLDM model degenerates into SVM. CDLDM enhances the architecture of SVM by optimizing the marginal distribution and extending it to cross-domain domains. Specifically, the CDLDM model aims to improve the generalization ability and performance of the model by optimizing the marginal distribution. By minimizing feature differences between different domains, the model's adaptability in different domains is further enhanced. Overall, these improvements enable the model to better adapt to different domains of data and enhance its performance in practical applications.

Incorporating Eqs. (1) and (2) into Eq. (5), we can derive the quadratic programming problem as follows

$$\begin{aligned} \min_{\omega, \xi} f = & \frac{1}{2} \omega^T \omega + \frac{2\lambda_1}{m^2} (m\omega^T \phi(\mathbf{X}_S) \phi(\mathbf{X}_S)^T \omega \\ & - \omega^T \phi(\mathbf{X}_S) \mathbf{Y}_S \mathbf{Y}_S^T \phi(\mathbf{X}_S)^T \omega) - \lambda_2 \frac{1}{m} (\phi(\mathbf{X}_S) \mathbf{Y}_S)^T \omega \\ & + C \sum_{i=1}^m \xi_i + \lambda D_g(\mathbf{X}_S, \mathbf{X}_T), \quad (7) \\ s.t. \ y_i \omega^T \phi(\mathbf{x}_i) \geq & 1 - \xi_i, \\ \xi_i \geq 0, \quad i = & 1, 2, \dots, m. \end{aligned}$$

Eq. (7) is usually difficult to handle. According to the representation theorem in Belkin et al. (2006), we can rephrase Eq. (7) using the following Representer theorem.

Theorem 1. *The optimal solution ω of Eq. (7) can be succinctly expressed as*

$$\omega = \phi(X_{st})\alpha = \sum_{i=1}^m \alpha_i \phi(x_i) + \sum_{j=1}^n \alpha_{j+m} \phi(z_j) \quad (8)$$

where $x_i \in X_S$, $z_j \in X_T$, $X_{st} = (x_{i=1}^m, z_{j=1}^n)$, and $\alpha = [\alpha_1, \dots, \alpha_m, \dots, \alpha_{m+n}]^T$ are the specific coefficients.

Proof. Due to the correlation between the projection vector ω and both the source and target samples, ω can be decomposed into a part that exists within the span of $\phi(x_i)$ and $\phi(z_j)$ and an orthogonal part vector, i.e.,

$$\omega = \sum_{i=1}^m \alpha_i \phi(x_i) + \sum_{j=1}^n \alpha_{j+m} \phi(z_j) + \beta = \phi(X_{st})\alpha + \beta, \quad (9)$$

where β denotes a vector that adheres to the condition $\phi(X_{st})\beta = 0$ [Zhang & Zhou \(2014b\)](#). Then, we have

$$\phi(X_S)^T \omega = \phi(X_S)^T (\phi(X_{st})\alpha + \beta) = \phi(X_S)^T \phi(X_{st})\alpha, \quad (10)$$

so the second and the third terms in the (7) are unrelated to β , and the fourth term is independent of β . Additionally, the constraint is also not correlated with β . For the first term and last terms related to β in the (7), according to $\phi(X_{st})\beta = 0$, we can obtain (only the first term is extrapolated here, and the last term is the same)

$$\begin{aligned} \omega^T \omega &= (\phi(X_{st})\alpha + \beta)^T (\phi(X_{st})\alpha + \beta) \\ &= \alpha^T \phi(X_{st})^T \phi(X_{st})\alpha + \beta^T \beta \\ &\geq \alpha^T \phi(X_{st})^T \phi(X_{st})\alpha, \end{aligned} \quad (11)$$

where if and only if $\beta = 0$, the equal sign holds.

Therefore, the β setting to 0 does not influence the second, third, and fourth terms, but strictly reduces the first and last items of (7). Hence, Theorem 1 is accepted.

3.2.2. The measure of projection distribution discrepancy

In this section, we provide a detailed introduction to the projection marginal distribution discrepancy D_f , the projection scattering distribution discrepancy S_f , and the graph-based Laplacian regularization P_f .

1) Projected Marginal Distribution Discrepancy

In the context of E-noses drift issues, our primary objective is to mitigate the differences in data distributions across various domains by means of learning a feature transformation. Consequently, the effective measurement of distribution disparities between domains becomes a paramount concern. In this pursuit, it is crucial to select a method suitable for quantifying distribution dissimilarity, while ensuring that the chosen method is computationally efficient and easily integrated into the optimization problem of LDM. Maximum Mean Discrepancy (MMD) has been demonstrated as an effective approach for estimating the distance between two distributions, offering a relatively straightforward implementation with fast computation speed. Therefore, we utilize the MMD metric to measure the distribution

differences. The D_f is defined as

$$\begin{aligned}
D_f(X_S, X_T) &= \omega^T \left\| \frac{1}{m} \sum_{i=1}^m \phi(x_i) - \frac{1}{n} \sum_{j=1}^n \phi(z_j) \right\|_{\mathcal{H}}^2 \omega \\
&= \left\| \frac{1}{m} \sum_{k=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{i=1}^m \phi(x_i) - \frac{1}{n} \sum_{k=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{j=1}^n \phi(z_j) \right\|^2 \\
&= \frac{1}{m^2} \sum_{k,l=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{i=1}^m \phi(x_i) \sum_{j=1}^m \phi(x_j) \phi(x_l) \alpha_l \\
&\quad + \frac{1}{n^2} \sum_{k,l=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{i=1}^n \phi(z_i) \sum_{j=1}^n \phi(z_j) \phi(x_l) \alpha_l \\
&\quad - \frac{1}{mn} \left(\sum_{k,l=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{i=1}^m \phi(x_i) \sum_{j=1}^n \phi(z_j) \phi(x_l) \alpha_l \right. \\
&\quad \left. - \sum_{k,l=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{j=1}^n \phi(z_j) \sum_{i=1}^m \phi(x_i) \phi(x_l) \alpha_l \right) \\
&= \alpha^T \left(\frac{1}{n^2} K_S[1]^{m \times m} K_S^T + \frac{1}{m^2} K_T[1]^{n \times n} K_T^T \right. \\
&\quad \left. - \frac{1}{nm} (K_S[1]^{m \times n} K_T^T + K_T[1]^{n \times m} K_S^T) \right) \alpha \\
&= \alpha^T \Omega_d \alpha,
\end{aligned} \tag{12}$$

where $[1]^{m \times m}$ represents a $m \times m$ matrix of all ones. $K_S \in \mathbb{R}^{(m+n) \times m}$ and $K_T \in \mathbb{R}^{(m+n) \times n}$ is the kernel matrix for the source samples and the target samples, respectively. Ω_d is a $(m+n) \times (m+n)$ symmetrical positive semi-definite kernel matrix, which is defined as

$$\begin{aligned}
\Omega_d &= \frac{1}{n^2} K_S[1]^{m \times m} K_S^T + \frac{1}{m^2} K_T[1]^{n \times n} K_T^T \\
&\quad - \frac{1}{nm} (K_S[1]^{m \times n} K_T^T + K_T[1]^{n \times m} K_S^T).
\end{aligned} \tag{13}$$

2) Projection Scattering Distribution Discrepancy

In order to effectively assess the distribution differences between the source domain and the target domain, we need to consider the variance or dispersion of the data. This is particularly important because dispersion reflects the degree of spread of data points within the distribution, providing crucial information about the breadth and tightness of the data distribution. Therefore, it is necessary to make full use of the scatter information from source and target domains

in order to comprehensively assess distribution differences. The S_f is defined as

$$\begin{aligned}
 S_f(X_S, X_T) &= \omega^T \left| \frac{1}{m} \sum_{i=1}^m \phi(x_i) \phi(x_i)^T - \frac{1}{n} \sum_{j=1}^n \phi(z_j) \phi(z_j)^T \right|_{\mathcal{H}} \omega \\
 &= \left| \frac{1}{m} \sum_{j,k}^{m+n} \alpha_j^T \phi(x_j)^T \sum_{i=1}^m \phi(x_i) \phi(x_i)^T \alpha_k \phi(x_k) \right. \\
 &\quad \left. - \frac{1}{n} \sum_{j,k}^{m+n} \alpha_j^T \phi(x_j)^T \sum_{i=1}^n \phi(z_i) \phi(z_i)^T \alpha_k \phi(x_k) \right| \\
 &= \left| \frac{1}{m} \sum_{j,k}^{m+n} \alpha_j^T \alpha_k \sum_{i=1}^m k(x_j, x_i) k(x_i, x_k) \right. \\
 &\quad \left. - \frac{1}{n} \sum_{j,k}^{m+n} \alpha_j^T \alpha_k \sum_{i=1}^n k(x_j, z_i) k(z_i, x_k) \right| \\
 &= \alpha^T \left| \frac{1}{m} K_S K_S^T - \frac{1}{n} K_T K_T^T \right| \alpha = \alpha^T \Omega_s \alpha,
 \end{aligned} \tag{14}$$

where Ω_s is a $(m+n) \times (m+n)$ symmetrical positive semi-definite kernel matrix, which is defined as

$$\Omega_s = \left| \frac{1}{m} K_S K_S^T - \frac{1}{n} K_T K_T^T \right|. \tag{15}$$

3) Graph-based Laplacian regularization

Manifold theory suggests that samples near each other in the original sample space will also be close in the projected RKHS, indicating a higher likelihood of belonging to the same class. Thus, in the context of E-nose data analysis, it is reasonable to consider that samples that are close together in the original feature space are probably from the same class. In order to maintain the geometric relationship between samples in RKHS, we added a graph-based Laplacian regularization term to the optimization function of CDLDM. The P_f is defined as

$$\begin{aligned}
 P_f(X_S, X_T) &= \omega^T \left\| \sum_{i,j=1}^{m+n} W_{ij} (f(x_i) - f(x_j))^2 \right\|_{\mathcal{H}} \omega \\
 &= \sum_{k=1}^{m+n} \alpha_k^T \phi(x_k)^T \sum_{i,j=1}^{m+n} \phi(x_i) L_{ij} \phi(x_j) \sum_{l=1}^{m+n} \alpha_l \phi(x_l) \\
 &= \alpha^T K_{st} L K_{st}^T \alpha \\
 &= \alpha^T \Omega_p \alpha,
 \end{aligned} \tag{16}$$

where $K_{st} \in \mathbb{R}^{(m+n) \times (m+n)}$ is the kernel matrix for samples in X_{st} , $\Omega_p = K_{st} L K_{st}^T$. W is the proximity relationship matrix between samples, which is defined as

$$W_{ij} = \begin{cases} 1 & x_i \in N_p(x_j) \text{ or } x_j \in N_p(x_i) \\ 0 & \text{otherwise} \end{cases}, \tag{17}$$

where $N_p(x_i)$ is the sets of the p -nearest neighbors of x_i . L is the graph Laplacian matrix and is computed as $L = D - W$, D is a diagonal matrix with $D_{ij} = \sum_{j=1}^{m+n} W_{ij}$.

3.2.3. Model Optimization

In sections 3.2.2, we obtain the specific representation of the measurement of the projection distribution discrepancy in RHKS through D_f , S_f , and P_f , which can be further represented as

$$\begin{aligned} D_g &= \lambda_3 D_f + \lambda_4 S_f + \lambda_5 P_f \\ &= \alpha^T (\lambda_3 \Omega_d + \lambda_4 \Omega_s + \lambda_5 \Omega_p) \alpha \\ &= \alpha^T \Omega_g \alpha, \end{aligned} \quad (18)$$

where $\Omega_g = \lambda_3 \Omega_d + \lambda_4 \Omega_s + \lambda_5 \Omega_p$ is a $(n + m) \times (n + m)$ positive semi-definite kernel matrix.

Theorem 2. The dual formulation of the initial problem stated in Eq. (5) is expressed below

$$\begin{aligned} \min_{\boldsymbol{\eta}} f &= \frac{1}{2} \boldsymbol{\eta}^T \mathbf{M} \boldsymbol{\eta} + \left(\frac{\lambda_2 \mathbf{M} \mathbf{e} - m \mathbf{e}}{m} \right)^T \boldsymbol{\eta}, \\ \text{s.t. } &0 \leq \eta_i \leq C, \quad i = 1, \dots, m, \end{aligned} \quad (19)$$

Proof. Drawing from Theorem 1, we can infer the subsequent expressions

$$\begin{aligned} \omega^T \phi(X_S) &= \alpha^T \phi(X_{st})^T \phi(X_S) = \alpha^T \mathbf{K}_S, \\ \omega^T \omega &= \alpha^T \phi(X_{st})^T \phi(X_{st}) \alpha = \alpha^T \mathbf{G} \alpha, \end{aligned} \quad (20)$$

where $\mathbf{K}_S = \phi(X_{st})^T \phi(X_S)$ and $\mathbf{G} = \phi(X_{st})^T \phi(X_{st})$ are the kernel matrix. Let $\mathbf{K}_{:,i}$ represent the i -th column of $\mathbf{K}_S$. Substituting Eqs. (18) and (20) into Eq. (7), the matrix form of the CDLDM can be obtained

$$\begin{aligned} \min_{\alpha, \xi} f &= \frac{1}{2} \alpha^T \mathbf{Q} \alpha + \mathbf{q}^T \alpha + C \sum_{i=1}^m \xi_i, \\ \text{s.t. } &y_i \alpha^T \mathbf{K}_{:,i} \geq 1 - \xi_i, \quad i = 1, 2, \dots, m, \\ &\xi_i \geq 0, \quad i = 1, 2, \dots, m, \end{aligned} \quad (21)$$

where $\mathbf{Q} = 4\lambda_1(m\mathbf{K}_S\mathbf{K}_S^T - (\mathbf{K}_S\mathbf{Y}_S)(\mathbf{K}_S\mathbf{Y}_S)^T)/m^2 + \mathbf{G} + 2\Omega_g$ and $\mathbf{q} = -\lambda_2\mathbf{K}_S\mathbf{Y}_S/m$.

The Lagrangian formulation corresponding to the optimization problem delineated in Eq. (21) is articulated as follows

$$\begin{aligned} L(\alpha, \xi; \boldsymbol{\eta}, \boldsymbol{\delta}) &= \frac{1}{2} \alpha^T \mathbf{Q} \alpha + \mathbf{q}^T \alpha + C \sum_{i=1}^m \xi_i \\ &\quad - \sum_{i=1}^m \eta_i (\xi_i + y_i \alpha^T \mathbf{K}_{:,i} - 1) - \sum_{i=1}^m \delta_i \xi_i, \end{aligned} \quad (22)$$

where $\boldsymbol{\eta}$ and $\boldsymbol{\delta}$ are Lagrangian vectors. Let $\partial L / \partial \alpha = \partial L / \partial \xi_i = 0$, we obtain the following formulas

$$\begin{aligned} \frac{\partial L}{\partial \alpha} &= \mathbf{Q} \alpha + \mathbf{q} - \sum_{i=1}^m \eta_i y_i \mathbf{K}_{:,i} = 0, \\ \frac{\partial L}{\partial \xi_i} &= C - \eta_i - \delta_i = 0, \end{aligned} \quad (23)$$

introduced Eq. (23) into Eq. (22), the dual form of model (21) is exhibited as follow

$$\begin{aligned} \min_{\boldsymbol{\eta}} f &= \frac{1}{2} \boldsymbol{\eta}^T \mathbf{M} \boldsymbol{\eta} + \left(\frac{\lambda_2 \mathbf{M} \mathbf{e} - m \mathbf{e}}{m} \right)^T \boldsymbol{\eta}, \\ \text{s.t. } &0 \leq \eta_i \leq C, \quad i = 1, \dots, m, \end{aligned} \quad (24)$$

where $\mathbf{M} = [\text{diag}(\mathbf{Y}_S)] \mathbf{K}_S \mathbf{Q}^{-1} \mathbf{K}_S [\text{diag}(\mathbf{Y}_S)]$, $\mathbf{e} = [1, \dots, 1]_{1 \times m}$.

Based on Eq. (23), the parameter α can be determined using the subsequent expression

$$\begin{aligned}\alpha &= \mathbf{Q}^{-1} (\mathbf{K}_S[\text{diag}(\mathbf{Y}_S)]\boldsymbol{\eta} - \mathbf{q}) \\ &= \mathbf{Q}^{-1} \left(\frac{\lambda_2}{m} \mathbf{K}_S[\text{diag}(\mathbf{Y}_S)]\mathbf{e} + \mathbf{K}_S[\text{diag}(\mathbf{Y}_S)]\boldsymbol{\eta} \right) \\ &= \mathbf{Q}^{-1} \mathbf{K}_S[\text{diag}(\mathbf{Y}_S)] \left(\frac{\lambda_2 \mathbf{e} + m\boldsymbol{\eta}}{m} \right).\end{aligned}\quad (25)$$

Consequently, the classification of a novel test instance $\mathbf{z}$ can be carried out with the following decision function

$$f(\mathbf{z}) = \text{sgn}(\boldsymbol{\omega}^T \boldsymbol{\phi}(\mathbf{z})) = \text{sgn} \left(\sum_{i=1}^{m+n} \alpha_i k(\mathbf{x}_i, \mathbf{z}) \right). \quad (26)$$

3.3. The Algorithm and Complexity Analysis

For clarity, the algorithm pseudocode for the CDLDM model is explicitly delineated in Algorithm 1.

Algorithm 1: CDLDM

Input : samples $\mathbf{X}_S$ and $\mathbf{X}_T$, label $\mathbf{y}_S$, kernel matrix $\mathbf{K}_S$, $\mathbf{K}_T$ and $\mathbf{G}$.
Output: predict label of $\mathbf{X}_T$.
1 Calculate $\boldsymbol{\Omega}_d$, $\boldsymbol{\Omega}_s$ and $\boldsymbol{\Omega}_p$ by Eqs.(12), (14) and (16).
2 $\boldsymbol{\Omega}_g \leftarrow \text{Eq. (18)}$.
3 **while** Parameters are not exhaustively explored **do**
4 Update $\mathbf{C}$ and $\lambda_i (i = 1, 2, \dots, 5)$.
5 $\mathbf{Q} = 4\lambda_1(m\mathbf{K}_S\mathbf{K}_S^T - (\mathbf{K}_S\mathbf{Y}_S)(\mathbf{K}_S\mathbf{Y}_S)^T)/m^2$
6 $+ \mathbf{G} + 2\boldsymbol{\Omega}_g$.
7 $\mathbf{q} = -\lambda_2\mathbf{K}_S\mathbf{Y}_S/m$.
8 $\mathbf{M} = [\text{diag}(\mathbf{Y}_S)]\mathbf{K}_S\mathbf{Q}^{-1}\mathbf{K}_S[\text{diag}(\mathbf{Y}_S)]$.
9 **repeat**
10 Update variables $\boldsymbol{\eta}$ in Eq. (24).
11 **until** convergenc
12 $\alpha \leftarrow \boldsymbol{\eta}$ by Eq. (25).
13 **end while**
14 Obtain the optimal parameter combination to achieve the best model performance.
15 Compute the predicted label for the test sample $\mathbf{X}_T$ using Eq. (26).

4. Experiments

In this section, we undertake an extensive experimental assessment of the CDLDM across various sensor drift datasets. The details of the experimental setup are outlined in Section 4.1. We juxtapose the performance of CDLDM against other models across various experimental configurations in Section 4.2. In Section 4.3, we conduct experiments on a more complex dataset. Section 4.4 is dedicated to an analysis of how sensitive the CDLDM model is to changes in its parameters. Finally, in order to analyze the roles of various components of CDLDM more clearly, we conducted ablation experiments in section 4.5.

4.1. Experimental Settings

To demonstrate the superiority of the CDLDM model, we choose two classic dimensionality reduction methods, i.e. principal component analysis (PCA) and linear discriminant analysis (LDA), for comparison. Additionally, we utilize five semi-supervised learning techniques that are based on manifold learning: neighborhood component analysis (NCA) Goldberger et al. (2004), locality preservation projection (LPP) He & Niyogi (2003), neighborhood preserving embedding (NPE) He et al. (2005), local fisher discriminant analysis (LFDA) Sugiyama (2006), and multidimensional scaling (MDS) Torgerson & S (1952). In addition, we also select DAKSVM (Tao et al., 2012) and subspace learning

Table 1

Parameters settings.

Parameters	The range of validation
C	$\{2^{-4}, 2^{-2}, 2^0, 2^2, 2^4\}$
λ_1, λ_2	$\{10^{-4}, 10^{-2}, 10^0, 10^2, 10^4\}$
$\lambda_3, \lambda_4, \lambda_5$	$\{10^{-6}, 10^{-4}, \dots, 10^4, 10^6\}$
γ	$\{2^{-4}, 2^{-2}, 2^0, 2^2, 2^4\}$
p	$\{5, 10, 20, 50, 100, 200\}$

p is the parameter in Eq. (17), representing the nearest neighbor quantity. γ is the parameter in the RBF kernel function.

Table 2

Details of UCSD dataset.

Batch ID	Month	C ₃ H ₆ O	C ₂ H ₄ O	C ₂ H ₅ OH	C ₂ H ₄	NH ₃	C ₇ H ₈	Total
Batch 1	1,2	90	98	83	30	70	74	445
Batch 2	3~10	164	334	100	109	532	5	1244
Batch 3	11,12,13	365	490	216	240	275	0	1586
Batch 4	14,15	64	43	12	30	12	0	161
Batch 5	16	28	40	20	46	63	0	197
Batch 6	17~20	514	574	110	29	606	467	2300
Batch 7	21	649	662	360	744	630	568	3613
Batch 8	22,23	30	30	40	33	143	18	294
Batch 9	24,30	61	55	100	75	78	101	470
Batch 10	36	600	600	600	600	600	600	3600

Table 3UCSD dataset classification accuracies under **Setting 1** (%).

Algorithms	Batch ID								Average
	5 → 1	5 → 2	5 → 3	5 → 4	9 → 6	9 → 7	9 → 8	9 → 10	
DRCA	48.32	72.59	53.34	76.40	42.48	33.77	82.89	30.75	55.07
LPP	41.57	66.08	81.21	90.68	67.96	50.29	87.76	45.72	66.41
MDS	42.02	71.30	67.21	55.28	44.22	28.76	59.52	24.17	49.06
NCA	46.07	79.26	68.66	70.81	34.33	28.43	75.85	31.33	54.34
NPE	39.55	65.19	90.21	88.82	53.39	50.65	89.12	40.28	53.37
LFDA	39.33	67.36	74.91	78.26	64.61	62.22	89.80	47.39	65.48
PCA	42.02	68.41	68.16	70.81	59.91	35.37	68.37	32.25	55.66
LDA	50.56	73.87	82.35	86.96	55.26	49.54	87.76	46.03	66.54
DAKSVM	42.02	75.48	71.88	77.27	77.83	47.55	70.41	52.82	64.41
CDLDM	75.28	91.72	96.28	84.47	81.96	56.05	88.10	58.11	79.00

method domain regularized component analysis (DRCA) [Zhang et al. \(2017\)](#). CDLDM mainly involves eight adjustable parameters, and the range of parameter traversal during the experiment is detailed in Table 1.

To appraise the anti-drift capabilities of CDLDM across different temporal scales, we have instituted two experimental settings, which are detailed as follows:

Setting 1: The training dataset is fixed, with batches five and nine serving as source domain data, and all other batches as target domain data.

Setting 2: The training dataset undergoes a continuous update process in which batch k is used for the source domain, and the following batch, $(k + 1)$, is used as the target domain.

Subsequently, these models are executed under two distinct settings, but their performance is assessed under a consistent setting to enable fair comparison. The accuracy of each algorithm is contingent upon the optimal configuration of its parameters.

Table 4UCSD dataset classification accuracies under **Setting 2** (%).

Algorithms	Batch ID									Average
	1 → 2	2 → 3	3 → 4	4 → 5	5 → 6	6 → 7	7 → 8	8 → 9	9 → 10	
DRCA	77.49	81.43	70.19	84.77	43.55	72.05	69.39	62.34	32.72	65.99
LPP	63.76	75.73	77.02	73.10	52.22	83.42	76.87	65.32	45.67	68.12
MDS	75.48	53.72	78.26	72.59	55.87	50.12	31.29	48.55	27.00	54.76
NCA	69.94	67.65	79.50	86.80	55.35	61.08	44.56	56.81	28.50	61.13
NPE	45.02	92.94	80.75	97.97	75.00	84.61	82.31	70.21	44.92	74.86
LFDA	42.23	61.29	83.85	66.50	52.48	70.41	58.57	61.49	42.11	59.88
PCA	36.98	60.91	65.84	98.48	57.65	48.96	38.10	44.89	32.19	53.78
LDA	69.13	70.62	57.14	97.97	59.48	66.79	77.89	68.09	46.03	68.13
DAKSVM	82.15	64.75	55.28	60.46	52.35	58.82	66.46	44.68	51.19	59.57
CDLDM	82.64	90.61	90.68	94.42	75.61	71.91	82.31	72.77	64.56	80.61

Table 5

Details of CQU dataset.

Batch	DoF	HCHO	C ₆ H ₆	C ₇ H ₈	CO	NO ₂	NH ₃	Total
Master	6	126	72	66	58	38	60	420
Slave 1	6	108	108	106	98	107	81	608
Slave 1	6	108	87	94	95	108	84	576

4.2. Experiment on sensor drift dataset UCSD

To ascertain the efficacy of the CDLDM model, we have leveraged a three-year benchmark dataset concerning sensor drift, which was collected by Vergara et al. at UCSD [Vergara et al. \(2012b\)](#). This dataset meticulously documents a total of 13,910 measurements, amassed from January 2008 to February 2011, on natural gas transmission platforms. The data were systematically gathered by an E-nose system, which is comprised of an ensemble of 16 metal-oxide semiconductor gas sensors. These sensors are capable of discerning six distinct concentrations of pure gaseous compounds, specifically acetaldehyde, acetone, ammonia, ethylene, toluene and ethanol. Each sensor within the system is engineered to extract a comprehensive set of eight features, culminating in the generation of a high-dimensional feature vector of 128 dimensions for every sample (16 sensors multiplied by 8 features). For an in-depth exploration of the methodology employed to ascertain these eight features per sensor, the interested reader is directed to the scholarly work of [Vergara et al. \(2012b\)](#). A comprehensive overview of the dataset's characteristics is delineated in Table 2.

In this section, two sets of experiments are performed using the aforementioned dataset — one under Setting 1 and the other under Setting 2.

1) Table 3 records the results of each algorithm under setting 1. The findings highlight several key observations:

- Across numerous batches, the CDLDM model consistently achieves the highest accuracy, showcasing its superior resistance to sensor drift.
- It is worth noting that although CDLDM did not reach its maximum value in batches 4, 7, and 8, it still demonstrated strong performance compared to other algorithms.
- Among all algorithms, some algorithms have shown particularly good performance in certain batches, for example, LPP has the highest accuracy in batch 4, reaching 90.68%. However, when considering the overall performance of the algorithm, CDLDM shows the highest average accuracy, up to 79%, far higher than other algorithms, highlighting its significant advantages over other algorithms.

2) Table 4 records the results of each algorithm under setting 2. The data drawn from these experiments yield several notable observations:

Table 6
CQU dataset classification accuracies.

Task	DRCA	LPP	MDS	NCA	NPE	LFDA	PCA	LDA	DAKSVM	CDLDM
Master $\rightarrow$ slave1	49.01	39.14	40.30	51.81	56.25	48.03	46.71	49.34	46.27	56.91
Master $\rightarrow$ slave2	53.65	51.39	40.97	53.13	56.07	46.35	40.97	47.57	53.28	62.47
Average	51.33	45.27	40.63	52.47	56.16	47.19	43.84	48.46	49.78	59.69

- Similar to Setting 1, CDLDM is a leading algorithm in almost all batches. This emphasizes its robust performance in anti-drift performance, whether in scenarios where the training set dynamically changes with each batch update or when the training set remains unchanged.
- We can notice that some algorithms perform better under Setting 2, because the shorter the interval between two batches, the closer the data distribution. For example, compared to Setting 1, the average accuracy of NPE increased by 21.49%, and it achieved the maximum value in batches 3, 7, and 8. But its overall performance is characterized by fluctuations across batches, indicating that NPE may perform well in specific scenarios, but may not always be superior to other algorithms. However, CDLDM demonstrated significant performance in both settings, demonstrating its robustness to constantly changing environmental conditions.

Overall, CDLDM exhibits excellent and stable performance in both settings. Compared to Setting 1, Setting 2 improves the performance of most algorithms by dynamically changing the training set to make the distribution of the source and target domains closer together. Specifically, CDLDM emerges as the algorithm that attains the highest mean accuracy across both experimental settings, at 79% and 80.61% respectively, far higher than other algorithms, demonstrating its excellent anti-drift ability. In Setting 2, due to the continuous updating of the training set, the distribution between the source domain and the target domain is closer, which provides more favorable conditions for each algorithm. Therefore, most algorithms have relatively improved performance under Setting 2, especially with the most significant improvement in NPE.

4.3. Experiment on sensor drift dataset CQU

An examination of a complex E-noses dataset is undertaken in this section. The dataset, curated by Zhang et al. [Zhang & Tian \(2014\)](#) at CQU, encompasses the detection of six distinct gases at varying concentrations: ammonia (NH₃), toluene (C₇H₈), formaldehyde (HCHO), carbon monoxide (CO), benzene (C₆H₆), and nitrogen dioxide (NO₂). The dataset harnesses data from three E-noses systems: a primary system (source domain) and two slave systems (target domain) with a five-year gap between their respective data collection periods. Each system is equipped with four gas sensors and additional modules for temperature and humidity sensing. As a result, each experimental sample yields a 6-dimensional feature vector. Comprehensive details of the dataset are provided in Table 5.

The dataset introduced by Zhang et al. [Zhang & Tian \(2014\)](#) from CQU presents a higher level of complexity and exhibits more pronounced distributional variations when contrasted with the UCSD dataset. It comprises source domain data from a master system and two target domain datasets from slave systems. To ascertain the robustness of the CDLDM in scenarios with substantial distributional differences between source and target domains, we conduct comparative experiments with other models. Table 6 shows the experimental results with accuracy as the indicator. CDLDM achieves optimal results in both target domains and is far superior to other algorithms in slave2.

4.4. Experiment on parameter sensitivity

To further evaluate the stability of CDLDM, we conducted parameter sensitivity analysis under setting 1 using the UCSD dataset in this section. CDLDM has six pivotal parameters: C , λ_1 , λ_2 , λ_3 , λ_4 , λ_5 . We employ the grid search method to determine the optimal values for each parameter. The specific experimental results are shown in Fig. 2.

Fig. 2 depicts the accuracy variation trends of various parameters of the CDLDM algorithm on different batches. We observe that C and λ_2 consistently exhibit stable accuracy across the majority of batches. Meanwhile, λ_3 demonstrates relative stability when its value is large, whereas λ_4 maintains relative stability when its value is small. λ_1 and λ_5 are relatively volatile, but they can still achieve optimal values across a wide range of values. Based on these findings, we can efficiently identify the optimal value ranges for each parameter, which will facilitate our subsequent research.

Table 7UCSD dataset ablation classification accuracies under **Setting 1** (%).

case	Batch ID								Average
	5 $\rightarrow$ 1	5 $\rightarrow$ 2	5 $\rightarrow$ 3	5 $\rightarrow$ 4	9 $\rightarrow$ 6	9 $\rightarrow$ 7	9 $\rightarrow$ 8	9 $\rightarrow$ 10	
Scenario 1	41.80	55.23	39.85	63.35	22.70	17.96	70.07	18.28	44.42
Scenario 2	56.18	67.77	53.91	83.23	30.96	35.18	58.50	32.39	55.10
Scenario 3	52.58	81.27	98.36	61.49	47.39	23.64	69.05	38.22	61.97
Scenario 4	59.55	74.76	96.53	80.12	62.87	45.20	66.33	47.19	69.34
Scenario 5	71.46	77.57	98.11	83.85	43.48	28.98	72.11	32.53	67.94
Scenario 6	60.00	80.55	98.23	80.75	65.96	53.89	86.44	43.78	75.12
Scenario 7	64.04	85.77	97.79	79.50	61.91	49.40	78.23	47.36	73.81
CDLDM	75.28	91.72	96.28	84.47	81.96	56.05	88.10	58.11	79.00

4.5. Ablation study

In this section, we performed ablation experiments to illustrate that CDLDM substantially enhances the capability of LDM in cross-domain learning. A total of seven distinct scenarios require comparison:

Scenario 1: By setting all parameters λ_5 , λ_4 , and λ_3 to 0, we comprehensively examined the performance of CDLDM in handling cross domain problems when all key components fail. At this scenario, the CDLDM model degenerates into the traditional LDM model, which helps to understand the overall response capability of the model when all key factors fail.

Scenario 2: By setting λ_4 and λ_3 to 0 simultaneously, we investigate the performance of CDLDM when the marginal distribution differences and the scattering distribution differences are weakened.

Scenario 3: Simultaneously setting λ_5 and λ_3 to 0, we investigate the performance of CDLDM while reducing attention to the marginal distribution differences and the graph-based Laplacian regularization.

Scenario 4: By simultaneously setting λ_5 and λ_4 to 0, we delve into the performance of CDLDM when the scattering distribution differences and the graph-based Laplacian regularization are both weakened. This helps to reveal the overall adaptability of the model when the trade-off parameters in these two aspects fail simultaneously.

Scenario 5: By setting λ_3 to 0, we completely eliminate the focus on differences in marginal distribution. The target of this scenario is to assess the adaptability of CDLDM to cross domain problems while ignoring differences in marginal distribution, with a focus on examining the role of λ_3 in the model.

Scenario 6: Setting λ_4 to 0 eliminates the focus on scattering distribution differences. This setting aims to investigate in depth the performance of CDLDM in handling cross domain problems while reducing attention to scattering distribution differences, highlighting the unique contribution of λ_4 in overall performance.

Scenario 7: We set λ_5 to 0. This means that we have excluded the influence of the trade-off parameter λ_5 in graph based Laplacian regularization. Through this setting, we aim to evaluate the ability of CDLDM to handle cross domain problems while ignoring geometric information between samples, emphasizing the role of λ_5 in the model.

Table 7 displays the results, revealing that omitting any component from the model results in diminished performance. Specifically, the more components of the model are lost, the overall trend of the model decreases. And when all components are removed, that is, when CDLDM degenerates into LDM, the performance of the model is the lowest, with an average accuracy of only 44.42%, which is 34.58% lower than the proposed model CDLDM. Furthermore, we can observe that when the projection marginal distribution discrepancy component of the model is removed, the performance of the model will be greatly affected. From the results, we can see that, excluding scenario 7, almost all batches corresponding to the lowest performance are missing the component of projection marginal distribution discrepancy. Overall, whether a certain part or two parts are missing, it will have an impact on the performance of the model, and our proposed model does greatly improve the cross-domain processing ability of traditional LDM.

5. Conclusion

This paper proposes a novel cross-domain large margin distribution machine, aiming to solve the challenges faced by traditional support vector machines when processing sensor drift data. CDLDM aims to minimize the distribution

discrepancy between source and target domains by simultaneously optimizing the projected maximum marginal distribution discrepancy, projected maximum distribution scattering discrepancy, and the manifold consistency under marginal distribution, thus achieving effective domain adaptation. In addition, CDLDM optimizes marginal distribution and achieves better generalization performance than SVM. An extensive array of systematic experiments conducted across diverse E-nose datasets has consistently underscored the exceptional cross-domain learning capabilities demonstrated by the CDLDM algorithm. Although CDLDM demonstrates strong cross-domain learning capabilities, it necessitates the optimization of a large number of parameters. In the future, we will focus on developing effective parameter optimization methods to achieve optimal results more efficiently. Overall, from the perspective of machine learning, we provide a new and efficient solution to the sensor drift problem. The proposed CDLDM is a domain adaptive classifier that has cross-domain capabilities and effectively solves the problem of distribution inconsistency caused by drift, providing a new perspective for exploring LDM theory.

CRedit authorship contribution statement

Junlin Chen: Methodology, Software, Writing - original draft. **Jinrui Guo:** Visualization, Data curation. **Wenting Hu:** Formal analysis, Software. **Wenzhe Pan:** Writing – review & editing. **Guojun Huang:** Funding acquisition, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the National Key Research and Development Program of China (No. 2021YFB3101500) and the Chongqing Municipal Training Program of Innovation and Entrepreneurship for Undergraduate (No. X202310635204).

References

- Belkin, M., Niyogi, P., & Sindhvani, V. (2006). Manifold regularization: A geometric framework for learning from labeled and unlabeled examples. *Journal of machine learning research*, 7.
- Di Carlo, S., Falasconi, M. et al. (2012). *Drift correction methods for gas chemical sensors in artificial olfaction systems: techniques and challenges*. INTECH Open Access Publisher London, UK. <https://doi.org/10.5772/22976>.
- Di Natale, C., Macagnano, A., Davide, F., D'Amico, A., Paolesse, R., Boschi, T., Faccio, M., & Ferri, G. (1997). An electronic nose for food analysis. *Sensors and Actuators B: Chemical*, 44, 521–526. [https://doi.org/10.1016/S0925-4005\(97\)00175-5](https://doi.org/10.1016/S0925-4005(97)00175-5).
- Distante, C., Siciliano, P., & Vasanelli, L. (2000). Odor discrimination using adaptive resonance theory. *Sensors and Actuators B: Chemical*, 69, 248–252. Proceedings of the International Symposium on Electronic Noses.
- Dragonieri, S., Annema, J. T., Schot, R., van der Schee, M. P., Spanevello, A., Carratú, P., Resta, O., Rabe, K. F., & Sterk, P. J. (2009). An electronic nose in the discrimination of patients with non-small cell lung cancer and copd. *Lung Cancer*, 64, 166–170. <https://doi.org/10.1016/j.lungcan.2008.08.008>.
- Dragonieri, S., Schot, R., Mertens, B. J., Le Cessie, S., Gauw, S. A., Spanevello, A., Resta, O., Willard, N. P., Vink, T. J., Rabe, K. F., Bel, E. H., & Sterk, P. J. (2007). An electronic nose in the discrimination of patients with asthma and controls. *Journal of Allergy and Clinical Immunology*, 120, 856–862. <https://doi.org/10.1016/j.jaci.2007.05.043>.
- Dutta, R., Hines, E., Gardner, J., Kashwan, K., & Bhuyan, M. (2003). Tea quality prediction using a tin oxide-based electronic nose: an artificial intelligence approach. *Sensors and Actuators B: Chemical*, 94, 228–237. [https://doi.org/10.1016/S0925-4005\(03\)00367-8](https://doi.org/10.1016/S0925-4005(03)00367-8).
- Gao, W., & Zhou, Z.-H. (2013). On the doubt about margin explanation of boosting. *Artificial Intelligence*, 203, 1–18. <https://doi.org/10.1016/j.artint.2013.07.002>.
- Goldberger, J., Hinton, G. E., Roweis, S., & Salakhutdinov, R. R. (2004). Neighbourhood components analysis. In *Advances in Neural Information Processing Systems*. volume 17.
- Guo, T., Tan, X., Yang, L., Liang, Z., Zhang, B., & Zhang, L. (2022). Domain adaptive subspace transfer model for sensor drift compensation in biologically inspired electronic nose. *Expert Systems with Applications*, 208, 118237. <https://doi.org/10.1016/j.eswa.2022.118237>.
- Güney, S., & Atasoy, A. (2015). Study of fish species discrimination via electronic nose. *Computers and Electronics in Agriculture*, 119, 83–91. <https://doi.org/10.1016/j.compag.2015.10.005>.
- Haugen, J.-E., Tomic, O., & Kvaal, K. (2000). A calibration method for handling the temporal drift of solid state gas-sensors. *Analytica chimica acta*, 407, 23–39. [https://doi.org/10.1016/S0003-2670\(99\)00784-9](https://doi.org/10.1016/S0003-2670(99)00784-9).

- He, X., Cai, D., Yan, S., & Zhang, H.-J. (2005). Neighborhood preserving embedding. In *Tenth IEEE International Conference on Computer Vision (ICCV'05) Volume 1* (pp. 1208–1213 Vol. 2). volume 2. <https://doi.org/10.1109/ICCV.2005.167>.
- He, X., & Niyogi, P. (2003). Locality preserving projections. In *Advances in Neural Information Processing Systems*. volume 16.
- Hossein-Babaei, F., & Ghafarinia, V. (2010). Compensation for the drift-like terms caused by environmental fluctuations in the responses of chemoresistive gas sensors. *Sensors and Actuators B: Chemical*, 143, 641–648. <https://doi.org/10.1016/j.snb.2009.10.006>.
- Jiang, H., Xu, W., & Chen, Q. (2019). Evaluating aroma quality of black tea by an olfactory visualization system: Selection of feature sensor using particle swarm optimization. *Food Research International*, 126, 108605. <https://doi.org/10.1016/j.foodres.2019.108605>.
- Liu, B., Zeng, X., Tian, F., Zhang, S., & Zhao, L. (2019). Domain transfer broad learning system for long-term drift compensation in electronic nose systems. *IEEE Access*, 7, 143947–143959. <https://doi.org/10.1109/ACCESS.2019.2943188>.
- Liu, L., Chu, M., Gong, R., & Zhang, L. (2021). An improved nonparallel support vector machine. *IEEE Transactions on Neural Networks and Learning Systems*, 32, 5129–5143. <https://doi.org/10.1109/TNNLS.2020.3027062>.
- Long, M., Wang, J., Ding, G., Pan, S. J., & Yu, P. S. (2014a). Adaptation regularization: A general framework for transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 26, 1076–1089. <https://doi.org/10.1109/TKDE.2013.111>.
- Long, M., Wang, J., Ding, G., Pan, S. J., & Yu, P. S. (2014b). Adaptation regularization: A general framework for transfer learning. *IEEE Transactions on Knowledge and Data Engineering*, 26, 1076–1089. <https://doi.org/10.1109/TKDE.2013.111>.
- Marco, S., & Gutierrez-Galvez, A. (2012). Signal and data processing for machine olfaction and chemical sensing: A review. *IEEE Sensors Journal*, 12, 3189–3214. <https://doi.org/10.1109/JSEN.2012.2192920>.
- Martinelli, E., Magna, G., De Vito, S., Di Fuccio, R., Di Francia, G., Vergara, A., & Di Natale, C. (2013). An adaptive classification model based on the artificial immune system for chemical sensor drift mitigation. *Sensors and Actuators B: Chemical*, 177, 1017–1026. <https://doi.org/10.1016/j.snb.2012.11.107>.
- Padilla, M., Perera, A., Montoliu, I., Chaudry, A., Persaud, K., & Marco, S. (2010). Drift compensation of gas sensor array data by orthogonal signal correction. *Chemometrics and Intelligent Laboratory Systems*, 100, 28–35. <https://doi.org/10.1016/j.chemolab.2009.10.002>.
- Pardo, M., Niederjaufner, G., Benussi, G., Comini, E., Faglia, G., Sberveglieri, G., Holmberg, M., & Lundstrom, I. (2000). Data preprocessing enhances the classification of different brands of espresso coffee with an electronic nose. *Sensors and Actuators B: Chemical*, 69, 397–403. [https://doi.org/10.1016/S0925-4005\(00\)00499-8](https://doi.org/10.1016/S0925-4005(00)00499-8).
- Reyzin, L., & Schapire, R. E. (2006). How boosting the margin can also boost classifier complexity. (p. 753–760). <https://doi.org/10.1145/1143844.1143939>.
- Sain, S. R. (1996). The nature of statistical learning theory. *Technometrics*, 38, 409–409. <https://doi.org/10.1080/00401706.1996.10484565>.
- Sugiyama, M. (2006). Local fisher discriminant analysis for supervised dimensionality reduction. In *Proceedings of the 23rd international conference on Machine learning* (pp. 905–912). <https://dl.acm.org/doi/abs/10.1145/1143844.1143958>.
- Tao, J., Lai Chung, F., & Wang, S. (2012). On minimum distribution discrepancy support vector machine for domain adaptation. *Pattern Recognition*, 45, 3962–3984. <https://doi.org/10.1016/j.patcog.2012.04.014>.
- Torgerson, & S. W. (1952). Multidimensional scaling: I. theory and method. *Psychometrika*, 17, 401–419. <https://doi.org/10.1007/BF02288916>.
- Vergara, A., Vembu, S., Ayhan, T., Ryan, M. A., Homer, M. L., & Huerta, R. (2012a). Chemical gas sensor drift compensation using classifier ensembles. *Sensors and Actuators B: Chemical*, 166–167, 320–329. <https://doi.org/10.1016/j.snb.2012.01.074>.
- Vergara, A., Vembu, S., Ayhan, T., Ryan, M. A., Homer, M. L., & Huerta, R. (2012b). Chemical gas sensor drift compensation using classifier ensembles. *Sensors and Actuators B: Chemical*, 166–167, 320–329. <https://doi.org/10.1016/j.snb.2012.01.074>.
- Wang, L., Sugiyama, M., Jing, Z., Yang, C., Zhou, Z.-H., & Feng, J. (2011). A refined margin analysis for boosting algorithms via equilibrium margin. *The Journal of Machine Learning Research*, 12, 1835–1863. <https://www.jmlr.org/papers/volume12/wang11a/wang11a.pdf>.
- Yan, J., Chen, F., Liu, T., Zhang, Y., Peng, X., Yi, D., & Duan, S. (2022). Subspace alignment based on an extreme learning machine for electronic nose drift compensation. *Knowledge-Based Systems*, 235, 107664. <https://doi.org/10.1016/j.knosys.2021.107664>.
- Zhang, L., Liu, Y., He, Z., Liu, J., Deng, P., & Zhou, X. (2017). Anti-drift in e-nose: A subspace projection approach with drift reduction. *Sensors and Actuators B: Chemical*, 253, 407–417. <https://doi.org/10.1016/j.snb.2017.06.156>.
- Zhang, L., & Tian, F. (2014). Performance study of multilayer perceptrons in a low-cost electronic nose. *IEEE Transactions on Instrumentation and Measurement*, 63, 1670–1679. <https://doi.org/10.1109/TIM.2014.2298691>.
- Zhang, L., Tian, F., Liu, S., Dang, L., Peng, X., & Yin, X. (2013). Chaotic time series prediction of e-nose sensor drift in embedded phase space. *Sensors and Actuators B: Chemical*, 182, 71–79. <https://doi.org/10.1016/j.snb.2013.03.003>.
- Zhang, L., & Zhang, D. (2015). Domain adaptation extreme learning machines for drift compensation in e-nose systems. *IEEE Transactions on Instrumentation and Measurement*, 64, 1790–1801. <https://doi.org/10.1109/TIM.2014.2367775>.
- Zhang, T., & Zhou, Z.-H. (2014a). Large margin distribution machine. (p. 313–322). <https://doi.org/10.1145/2623330.2623710>.
- Zhang, T., & Zhou, Z.-H. (2014b). Large margin distribution machine. In *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 313–322).
- Zhang, T., & Zhou, Z.-H. (2018). Optimal margin distribution clustering. In *Proceedings of the AAAI Conference on Artificial Intelligence*. volume 32. <https://doi.org/10.1609/aaai.v32i1.11737>.
- Zuppa, M., Distante, C., Siciliano, P., & Persaud, K. C. (2004). Drift counteraction with multiple self-organising maps for an electronic nose. *Sensors and Actuators B: Chemical*, 98, 305–317. <https://doi.org/10.1016/j.snb.2003.10.029>.

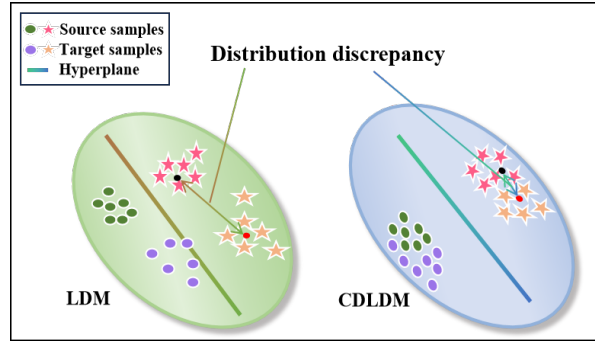


Fig. 1: CDLDM method schematic. LDM does not consider the distribution discrepancy of data, which makes it difficult for the classifier learned in the source domain to generalize effectively to the target domain. The proposed CDLDM aims to overcome this shortcoming of LDM. It is observed that when the distribution of source domain and target domain tends to be the same, the classifier learned from the source domain can be more effectively applied to the target domain.

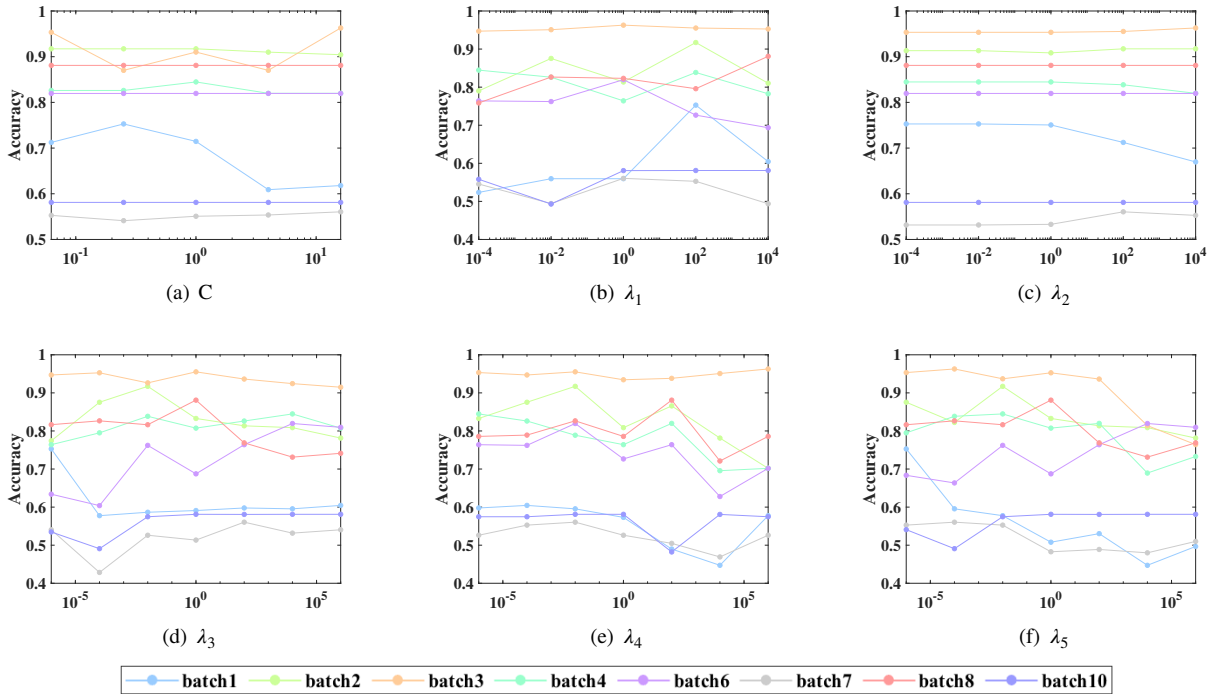


Fig. 2: Visualization of recognition accuracy of various parameters changing with batches.